Notes on Lagrange's equations,
MLS Chapter 4, Section 2

On eq. (4.3) p. 157

One should understand eq. (4.3) p. 157 as follows:

- The first matrix on the right hand side $\in \mathbb{R}^{3n\times 3n}$.
- The I post-multiplying m_1 and m_n (and by implication each m_i) is an identity matrix $\in \mathbb{R}^{3\times 3}$
- $\ddot{r}_1$ and $\ddot{r}_n$ (and by implication all $\ddot{r}_i$) are the acceleration vectors of the respective point masses, expressed in column vector form $\in \mathbb{R}^{3\times 1}$, just as in eq. (4.1)

On writing $\dot{R}r$ as $R\widehat{\omega}r$ on p. 162

In the second equation on p. 162 MLS substitute $\dot{R}r$ with $R\widehat{\omega}r$ without much explanation. Note that on the previous page, the velocity of point r is described in the inertial frame as $\dot{p} + \dot{R}r$. This is correct, as the coordinates of r in the inertial frame is $p + Rr$. The two terms in the velocity expression $\dot{p} + \dot{R}r$ can be interpreted as the velocity of the frame B origin $\dot{p}$ and the velocity of r relative to B, $\dot{R}r$, both of course in components in the inertial frame. The angular velocity of the body in the body frame is ω. Therefore, the velocity of r relative to B, expressed in the body frame, is $\widehat{\omega}r$. We can therefore say that $\dot{R}r = R\widehat{\omega}r$.

On eq. (4.9) p. 162

Revisit eq. (2.55) p. 55.